Introduction to Machine Learning

1. What Is Machine Learning?

Machine learning is a branch of artificial intelligence that enables computer systems to learn patterns from data and use those patterns to make predictions or decisions. Instead of explicitly programming every rule, we provide examples and allow an algorithm to estimate a useful relationship between inputs and outputs.

2. Types of Machine Learning

Supervised learning uses labeled examples. Common tasks include classification and regression. For example, a model can learn from historical house data to predict the price of a new house.

Unsupervised learning works with data without target labels. Clustering customers according to purchasing behavior is a common example.

Reinforcement learning learns through interaction with an environment. An agent receives rewards or penalties and gradually learns a policy for choosing actions.

The Machine Learning Workflow

3. From Data to Model

A typical machine learning project begins with data collection. The data is then cleaned, explored, transformed, and divided into training and evaluation sets. Features are selected or constructed to represent the information needed by the model.

During training, an algorithm adjusts its parameters to minimize an objective function. For supervised learning, the objective often measures the difference between predicted and actual target values.

4. Training and Evaluation

A model should not be judged only by how well it performs on training data. A separate validation or test set helps estimate how well the learned relationship generalizes to unseen examples. Metrics depend on the task: accuracy, precision, recall, and F1 score are frequently used for classification, while mean absolute error and mean squared error are common for regression.

Important Machine Learning Concepts

5. Overfitting and Underfitting

Overfitting occurs when a model learns the training examples too closely, including patterns that do not generalize to new data. A very complex model can have excellent training performance while performing poorly on unseen examples.

Underfitting occurs when a model is too simple to capture important relationships in the data. Both problems can be studied by comparing training and validation performance.

6. Feature Engineering

Feature engineering transforms raw information into representations that are useful for a learning algorithm. Examples include extracting the day of week from a timestamp, creating ratios between numerical variables, encoding categorical values, and scaling features when required by an algorithm.

7. Common Algorithms

Popular classical algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, k-nearest neighbors, and k-means clustering. The appropriate algorithm depends on the data, objective, computational constraints, and desired interpretability.